Calculus-II

Rolle's Theorem → If $f(x)$ is continuous in the closed interval $[a,b]$ and $f(x)$ exists in (a,b) and $f(a) = f(b)$, then there exist at least one value c $(a < c < b)$. Such that $f'(c) = 0$.

Summary:- (i) Continuous on $[a,b]$
(ii) Differentiable on (a,b) Such that,
(iii) $f(a) = f(b)$

there exists some (c) in (a,b), such that

- $f'(c) = 0$
- c lies between a,b. $[c \in (a,b)]$

Q.) ~~Veri~~ The function $f(x) = x(x+3)e^{-\frac{x}{2}}$ satisfy all conditions of Roll's mean value theorem in the interval $[-3,0]$. Then the value of c is:

$$f(x) = x(x+3)e^{-\frac{x}{2}}$$
$$f(x) = (x^2+3x)e^{-\frac{x}{2}}$$

$$f(-3) = ((-3)^2 + 3(-3))e^{-\frac{(-3)}{2}}$$
$$= (9-9)e^{3/2}$$
$$f(-3) = 0 \quad \text{— (i)}$$

$$f(0) = (0^2 + 3\times 0)e^{-0/2}$$
$$f(0) = 0 \quad \text{— (ii)}$$

from (i) and (ii)

$$f(-3) = f(0)$$

Now,

$$f(x) = (x^2+3x)\cdot e^{-\frac{x}{2}}$$

$$f'(x) = e^{-x/2}\left[2x+3\right] + (x^2+3x)\cdot e^{-x/2}\times\frac{-1}{2}$$

$$f'(x) = e^{-\frac{x}{2}}\left[(2x+3) - \left(\frac{x^2+3x}{2}\right)\right]$$

$$f'(x) = \frac{1}{e^{x/2}}\left[\frac{4x+6-x^2-3x}{2}\right]$$

$$f'(x) = \frac{1}{2e^{x/2}}\left[-x^2+x+6\right]$$

$$f'(c) = 0$$

$$\frac{1}{2e^{c/2}}\left[-c^2+c+6\right] = 0$$

$$-c^2+c+6=0$$
$$c^2-c-6=0$$
$$c^2-3c+2c-6=0$$
$$c(c-3)+2(c-3)=0$$
$$(c-3)(c+2)=0$$
$$c=3 \text{ or } c=-2$$
$$3 \notin (-3,0)$$

$$\therefore \boxed{c=-2}$$

→ Lagrange's Mean-Value Theorem :-

(i) Continuous on $[a,b]$
(ii) Differentiable on (a,b)

there exist some c in (a,b), such that

- $f'(c) = \dfrac{f(b) - f(a)}{b - a}$
- c lies between (a,b) $[c \in (a,b)]$

Q. Verify lagrange mean value theorem for the function $f(x) = \log_e x$ in $[1, e]$.

$f(x) = \log_e x$

$f(1) = \log_e 1$
$= 0$

$f(e) = \log_e e$
$= 1$

$f'(x) = \dfrac{1}{x}$

$f'(c) = \dfrac{1}{c}$

$f'(c) = \dfrac{f(e) - f(1)}{e - 1}$

$$\frac{1}{c} = \frac{1-0}{e-1}$$

$$c = e - 1 \qquad [e = 2.7 \text{ (appr)}]$$

$$\boxed{c = 1.7} \qquad \text{(appr)}$$

$$c \in (1, e)$$

Hence, verify lagrange mean value theorem

→ Cauchy's mean-value theorem :-

(i) $f(x)$ and $g(x)$ be continuous in $[a,b]$
(ii) $f'(x)$ and $g'(x)$ exist in (a,b)
(iii) $g'(x) \neq 0$ for any value of x in (a,b).

then there is at least one value c of x in (a,b), such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Note:- If $g(x) = x$ in $[a,b]$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

which is lagrange form.

Verify Cauchy's mean value theorem for the function

$\log_e x$ and $\frac{1}{x}$ in the interval $[1, e]$.

$f(x) = \log_e x$

$g(x) = \frac{1}{x}$

$f'(x) = \frac{1}{x}$

$g'(x) = -\frac{1}{x^2}$

both $f'(x)$ and $g'(x)$ exist in $(1, e)$

$f(1) = \log_e 1 = 0$

$f(e) = \log_e e = 1$

$g(1) = \frac{1}{1} = 1$

$g(e) = \frac{1}{e}$

$f'(c) = \frac{1}{c}$

$g'(c) = \cancel{-\frac{1}{x^2}} \quad -\frac{1}{c^2}$

$$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f(e)-f(1)}{g(e)-g(1)}$$

$$\frac{\frac{1}{c}}{\frac{-1}{c^2}} = \frac{1-0}{\frac{1}{e}-1}$$

$$-c = \frac{e}{1-e}$$

$$\boxed{c = \frac{e}{e-1}} \qquad \left[= \frac{2.7}{1.7} \text{ approx} \right]$$

$c \in (1, e)$.

Indeterminant form and L'Hospital's rule:-

* If $f(a) = \phi(a) = 0$

$$= \underset{x \to a}{Lt} \frac{f(a)}{\phi(a)} \quad \left[\frac{0}{0} \text{ form}\right]$$

$$= \underset{x \to a}{Lt} \frac{f'(a)}{\phi'(a)} \quad \left[\text{differentiating numberator and denominator separately}\right]$$

Then recheck if above result is also in $\left[\frac{0}{0} \text{ form}\right]$ then, repeat L'Hospitals rule:-

L'Hospitals rule:-

Eg:- $\underset{x \to 0}{Lt} \frac{\sin x}{x} \quad \left[\frac{0}{0}\right]$

after putting $x = 0$ then $\frac{0}{0}$ form we apply L'Hospital rule

$$\underset{x \to 0}{Lt} \frac{\sin' x}{x'} = \underset{x \to 0}{Lt} \frac{\cos x}{1} = 1$$

→ In determinant form :-

① $\frac{0}{0}$ ② $\frac{\infty}{\infty}$ ③ $0 \times \infty$

④ $\infty - \infty$ ⑤ 0^0 ⑥ 1^∞ ⑦ ∞^0

Rule :- change ②, ③, ④ in $\frac{0}{0}$ form or $\frac{\infty}{\infty}$ form and

Use L'Hospitals rule.

Eg :- $\underset{x \to 0}{Lt} \frac{\log x}{\cot x}$

putting $x = 0$ then $\frac{\infty}{\infty}$ form,

~~change it into $\frac{0}{0}$ form;~~

$$\cancel{\underset{x \to 0}{Lt} \frac{\log}{\frac{1}{\tan x}}}$$

$$\cancel{\underset{x \to}{Lt}}$$

$$= \underset{x \to 0}{Lt} \frac{(\log x)'}{(\cot x)'}$$

$$= \underset{x \to 0}{Lt} \frac{\frac{1}{x}}{-\operatorname{cosec}^2 x}$$

$$= \underset{x \to 0}{Lt} \frac{-\sin^2 x}{x} \qquad \left[\frac{0}{0} \text{ form}\right]$$

$$= -\underset{x \to 0}{Lt} \frac{(\sin^2 x)'}{x'}$$

$$= -\underset{x \to 0}{Lt} \frac{2\sin x \cos x}{1}$$

Now put $x = 0$ because no indeterminant form

$$= -2\sin 0 \cos 0 = 0$$

Eg: $\underset{x \to 0}{Lt} \tan x \log x$

putting $x = 0$ then, $0 \times \infty$ form

change it into $\frac{0}{0}$ or $\frac{\infty}{\infty}$

$$\underset{x \to 0}{Lt} \frac{\log x}{\cot x} \qquad \left[\tan x = \frac{1}{\cot x}\right]$$

$$\underset{x \to 0}{Lt} \frac{(\log x)'}{(\cot x)'}$$

$$\underset{x \to 0}{Lt} \frac{1/x}{-\operatorname{cosec}^2 x}$$

$$= -\underset{x \to 0}{Lt} \frac{\sin^2 x}{x} \qquad \left[\frac{0}{0} \text{ form}\right]$$

$$= \underset{x \to 0}{Lt} \frac{2\sin x \cos x}{1}$$

$$= 0$$

→ for (5), (6), (7)

0^0, 1^∞, ∞^0

* Use log;

eg; $\underset{x \to \pi/2}{Lt} (\sin x)^{\tan x}$

putting $x = \pi/2$ then [1^∞ form]

let $Y = \underset{x \to \pi/2}{Lt} (\sin x)^{\tan x}$

taking log both side

$$\log y = \underset{x \to \pi/2}{Lt} \log (\sin x)^{\tan x}$$

$$\log y = \underset{x \to \pi/2}{Lt} \left[\tan x \cdot \log \sin x \right]$$

$$\log y = \underset{x \to \pi/2}{Lt} \left[\frac{\log \sin x}{\cot x} \right] \quad \left[\frac{0}{0} \text{ form} \right]$$

$$\log y = \underset{x \to \pi/2}{Lt} \left[\frac{(\log \sin x)'}{(\cot x)'} \right]$$

$$= \underset{x \to \pi/2}{Lt} \left[\frac{\frac{1}{\sin x} \cdot \cos x}{-\operatorname{cosec}^2 x} \right]$$

$$= -\underset{x \to \pi/2}{Lt} \left[\frac{\sin^2 x \cdot \cos x}{\sin x} \right]$$

$$= -\underset{x \to \pi/2}{Lt} \left[\sin x \cos x \right]$$

$$\log y = 0$$

taking e both of side;

$$e^{\log y} = e^0$$

$$y = 1$$

$$\underset{x \to \pi/2}{Lt} (\sin x)^{\tan x} = 1$$

→ Maxima and Minima:-

- Process to find maxima and minima:-

Step:- 1- find $f'(x)$.

Step:- 2- Solve $f'(x) = 0$ to find the value of x.

Step:- 3- find $f''(x)$ and check

if;

- $f''(x) < 0$ (Negative), local maximum $f(x) = max^m$ at that value of x
- $f''(x) > 0$ (positive), local minimum $f(x) = min^m$ at that value of x
- $f''(x) = 0$ then, inflexion point

Eg; find max^m and min^m value of $3x^4 - 2x^3 - 6x^2 + 6x + 1$ at interval [0,2]

$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$

$f'(x) = 12x^3 - 6x^2 - 12x + 6$

$f'(x) = 0$

$12x^3 - 6x^2 - 12x + 6 = 0$

$2x^3 - x^2 - 2x + 1 = 0$

at $x = 1$

$$
\begin{array}{l}
\quad\quad\quad\quad\quad\quad 2x^2 + x - 1 \\
(x-1)\,\big)\; 2x^3 - x^2 - 2x + 1 \;\big(\\
\quad\quad\quad 2x^3 - 2x^2 \\
\quad\quad\quad\quad\quad + \\
\quad\quad\quad\quad x^2 - 2x \\
\quad\quad\quad\quad -x^2 + x \\
\quad\quad\quad\quad\quad -x + 1 \\
\quad\quad\quad\quad\quad -x + 1 \\
\quad\quad\quad\quad\quad\quad 0
\end{array}
$$

$(x-1)(2x^2 + x - 1) = 0$

$(x-1)(2x^2 + 2x - x - 1) = 0$

$(x-1)(2x(x+1) - 1(x+1)) = 0$

$(x-1)(x+1)(2x-1) = 0$

$x = 1$ or $x = -1$ or $x = 1/2$

$x = (1, 1/2) \in (0, 2)$

$x = (-1) \notin (0, 2)$

$f''(x) = 6x^2 - 2x - 2$

at $x = 1$

$f''(1) = 2$

at $x = 1/2$

$$f''(1/2) = 6 \times \frac{1}{4} - 2 \times \frac{1}{2} - 2$$

$$= \frac{3}{2} - 1 - 2$$

$$= \frac{3}{2} - 3$$

$$= -\frac{3}{2}$$

at $x = 1$, $f''(1) = 2$

$f''(x) > 0$

local minima
at $x = 1$

$$f(1) = 3(1)^4 - 2(1)^3 - 6(1)^2 + 6(1) + 1$$
$$= 2$$

The minimum value is $\underline{2}$

at $x = 1/2$, $f(1/2) = -3/2$

$f''(x) < 0$

local ~~max~~ maxima
at $x = 1/2$

$$f(1/2) = 3 \times \left(\frac{1}{2}\right)^4 - 2(1/2)^3 - 6(1/2)^2 + 6(1/2) + 1$$
$$= 39/16$$

$\therefore$ The maximum is $39/16$

Taylor's Theorem

$$f(x+a) = f(a) + \frac{x f'(a)}{1!} + \frac{x^2 f''(a)}{2!} + \frac{x^3 f'''(a)}{3!} + \cdots \frac{x^{n+1}}{(n+1)!} f^{n+1}(x+\theta a)$$

Remainder form
or
Error form
(R_n)

$$\Rightarrow \frac{x^{n+1}}{(n+1)!} f^{n+1}(x+\theta a)$$

$(0<\theta<1)$

Q) obtain the fourth-degree taylor's polynomial approximation to $f(x) = e^{2x}$ about $x=0$. Find the maximum error when $0 \le x \le 0.5$.

$\Rightarrow$ $f(x) = e^{2x}$

$f(x+0) = f(x)$

$a = 0$

$f(a) = e^{2a}$
$f(0) = e^{2\times 0} = e^0 = 1$
$f'(a) = 2e^{2a}$
$f'(0) = 2$
$f''(a) = 4e^{2a}$
$f''(0) = 4$
$f'''(a) = 8e^{2a}$
$f'''(0) = 8$

$f^{iv}(a) = 16e^{2a}$
$f^{iv}(0) = 16$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0)$$

$$+ \frac{x^4}{4!} f^{iv}(0)$$

$$= 1 + x2 + \frac{x^2}{2} 4 + \frac{x^3}{6} 8 + \frac{x^4}{24} 16$$

$$f(x) = 1 + 2x + 2x^2 + \frac{4x^3}{3} + \frac{2}{3} x^4$$

4th degree taylor's expansion of e^{2x}.

Maximum error;

$$R_4 = \frac{f^{4+1}(x+\theta a)}{(4+1)!} x^{4+1}$$

$$= \frac{x^5}{5!} f^5(x+\theta a)$$

$0 \le x \le 0.5$

$a = 0$

$$x \le \frac{(0.5)^5}{120} f^5(x)$$

$$x \le \frac{(0.5)^5}{120} \times 32e^{2x} \qquad [f^5(x) = 32e$$

$$x \le \frac{1}{2^5 \times 120} \times 32e^{2\times 0.5}$$

$$x \le \frac{1 \times 32 \times e^1}{32 \times 120}$$

$$x \le \frac{e}{120}$$

$$x \le 0.0266$$

Maximum error = 0.0266

Q. Using Taylor's theorem, express the polynomial $2x^3 + 7x^2 + x - 6$ in powers of $(x-1)$.

$\Rightarrow$ $f(x) = 2x^3 + 7x^2 + x - 6$

$$f(\underbrace{x-1}_{x} + \underbrace{1}_{a}) = f(x)$$

$x = x - 1$

$a = 1$

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \frac{(x-1)^4}{4!}f^{iv}(1) + \cdots$$

$f(x) = 2x^3 + 7x^2 + x - 6$

$f(1) = 4$

$f'(x) = 6x^2 + 14x + 1$

$f'(1) = 21$

$f''(x) = 12x + 14$

$f''(1) = 26$

$f'''(x) = 12$

$f'''(1) = 12$

$f^{iv}(x) = 0$

$f^{iv}(1) = 0$

$$f(x) = 4 + (x-1)21 + \frac{(x-1)^2}{2} 26 + \frac{(x-1)^3}{6} 12 + \frac{(x-1)^4}{24} 0 + \cdots$$

$$f(x) = 4 + 21(x-1) + 13(x-1)^2 + 2(x-1)^3$$

Taylor expansion of polynomial in powers of $(x-1)$

Q.1 Expand $\log x$ in powers of $(x-1)$ by taylor's theorem

$\Rightarrow$ $f(x) = \log x$

$$f(\underbrace{x-1}_{x} + \underbrace{1}_{a}) = f(x)$$

$x = x - 1$
$a = 1$

$$f(x) = f(1) + (x-1) f'(1) + \frac{(x-1)^2}{2!} f''(1) + \frac{(x-1)^3}{3!} f'''(1) + \frac{(x-1)^4}{4!} f^{iv}(1) + \cdots$$

$f(x) = \log x$

$f(1) = \log(1) = 0$

$f'(x) = \frac{1}{x}$

$f'(1) = 1$

$f''(x) = -\frac{1}{x^2}$

$f''(1) = -1$

$f'''(x) = \frac{2}{x^3}$

$f'''(1) = 2$

$f^{iv}(x) = \frac{-6}{x^4}$

$f^{iv}(1) = -6$

$$f(x) = 0 + (x-1)1 + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!} \times 2 + \frac{(x-1)^4}{4!}(-6) + \cdots$$

$$= x - 1 - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{6}$$

taylor expansion of $\log x$ in powers of $(x-1)$.

→ # Maclaurin's Theorem

putting $a = 0$, in taylor's series we get Maclaurin series;

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots \frac{x^{n+1}}{(n+1)!}f^{n+1}(\theta x)$$

Remainder form or error form (R_n) $\Rightarrow$ $\frac{x^{n+1}}{(n+1)!}f^{n+1}(\theta x)$

Q) If $f(x) = \log(1+x)$, $x > 0$, using Maclaurin theorem, show that for $0 < \theta < 1$.

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}$$

Deduce that $\log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3}$ for $x > 0$

$\Rightarrow$ Using Remainder form (R_2)

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(\theta x)$$

$f(x) = \log(1+x)$

$f(0) = \log(1+0) = 0$

$f'(x) = \frac{1}{1+x}$

$f'(0) = 1$

$f''(x) = \frac{-1}{(1+x)^2}$

$f''(0) = -1$

$f'''(x) = \frac{2}{(1+x)^3}$

$f'''(\theta x) = \frac{2}{(1+\theta x)^3}$

$$f(x) = 0 + x(1) + \frac{x^2}{2}(-1) + \frac{x^3}{3!}\frac{2}{(1+\theta x)^3}$$

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{(1+\theta x)^3}$$

$0 < \theta < 1$

$0 < x$

$x > 0, \theta > 0$

$\theta x > 0$

$1 + \theta x > 1$

$(1+\theta x)^3 > 1$

$$\frac{1}{(1+\theta x)^3} < 1$$

$$\frac{x^3}{(1+\theta x)^3} < x^3$$

$$\frac{x^3}{3(1+\theta x)^3} < \frac{x^3}{3}$$

$$x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3} < x - \frac{x^2}{2} + \frac{x^3}{3}$$

$$\log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3}$$